Criterion 8: Program Outcomes and Assessment

Sections 8.1 and 8.2: Outcome Statements, Alignment, and Course Design

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Based on `[2026-06-24] v2 SAR-CSE-criterion-8.pdf`, printed pp. 125–131.

What these two sections establish

8.1: Define the graduate outcomes

- State the twelve Program Outcomes (POs) expected at graduation.
- Map each PO to a Program Educational Objective (PEO).
- Explain why each relationship supports the program's objectives.

8.2: Translate outcomes into course work

- Develop and review measurable Course Outcomes (COs).
- Specify delivery, learning domains, and assessment tools.
- Link course-level learning to the relevant POs.

The alignment to explain

Program objectives ⟵ graduate outcomes ⟵ course outcomes ⟵ learning activities and assessment evidence.

Source: Criterion 8 SAR, Sections 8.1–8.2, pp. 125–131.

8.1: Technical capabilities expected at graduation

PO	Graduate attribute	What graduates should be able to do (summary)
PO1	Engineering knowledge	Apply mathematics, science, engineering fundamentals, and specialist knowledge to complex engineering problems.
PO2	Problem analysis	Identify, formulate, research, and analyze complex problems to reach substantiated conclusions.
PO3	Design/development of solutions	Design solutions meeting specified needs while considering health, safety, culture, society, and environment.
PO4	Investigation	Design experiments, analyze data, and synthesize research-based evidence into valid conclusions.
PO5	Modern tool usage	Select, create, and apply engineering and IT tools, including modeling, with awareness of their limitations.

Source: Criterion 8 SAR, Table 8.1, p. 125; statements condensed for presentation.

8.1: Professional capabilities expected at graduation

PO	Graduate attribute	What graduates should be able to do (summary)
PO6	The engineer and society	Assess societal, health, safety, legal, and cultural issues and professional responsibilities.
PO7	Environment and sustainability	Evaluate the societal and environmental impact and sustainability of engineering work.
PO8	Ethics	Apply ethical principles and uphold professional responsibilities and norms.
PO9	Individual work and teamwork	Work effectively independently and as a member or leader of diverse teams.
PO10	Communication	Write reports and design documents; present effectively; give and receive clear instructions.
PO11	Project management and finance	Apply engineering management and economic decision-making in individual and team projects.
PO12	Life-long learning	Recognize the need for and undertake independent, continuing learning as technology changes.

Source: Criterion 8 SAR, Table 8.1, pp. 125–126; statements condensed for presentation.

8.1: How the twelve POs support the four PEOs

PEO	Mapped POs	Reason for the alignment
PEO1	PO1–PO5	Computing knowledge, analysis, design, research, and modern tools support engineering solution development.
PEO2	PO6–PO8	Societal responsibility, sustainability, and ethics support responsible professional practice.
PEO3	PO12	Independent learning supports continuing professional development.
PEO4	PO9–PO11	Teamwork, communication, and management support collaboration in diverse and multidisciplinary settings.

What the mapping documents

Every PO has a stated connection to a PEO; the report supplies a justification for each connection. The PO approval document is referenced in Section G.1.

Source: Criterion 8 SAR, Table 8.2 and mapping justifications, pp. 126–127; PEO descriptions above summarize the rationale, not official PEO wording.

8.2: How course outcomes are developed and reviewed

1. **Instructors draft the course package:** outline, contents, assessment plan, and CO–PO mappings using the approved OBE curriculum and syllabus.
2. **Course coordinators harmonize sections:** align contents, assessment, and mappings across multiple sections of a course.
3. **The Departmental Committee reviews:** examine proposed materials, previous attainment, CQI findings, and instructor, student, alumni, and employer feedback.
4. **The Syllabus Committee examines curricular changes:** check alignment with the mission, POs, and PEOs; significant changes follow institutional approval through the Academic Council.
5. **The department finalizes the COs:** incorporate approved improvements into outcomes, outlines, and assessment strategies.

The recurring work

Use evidence and feedback from earlier offerings to improve the next course design.

Source: Criterion 8 SAR, Section 8.2, p. 127.

8.2.1: What is documented for each course outcome?

The CO specification

- A specific, measurable statement of expected knowledge, skills, or competence.
- One or more mapped POs.
- Learning-domain codes.
- Delivery methods and learning activities.
- Assessment tools that can demonstrate the outcome.

Example: AI theory, CO3

Student work: recommend solutions to real-life AI problems using existing techniques.

Alignment: PO3 (design).
Domain: C6.
Delivery: lecture and demonstration.
Evidence: mid-term and final examinations.

Domain families: C = cognitive; A = affective; P = psychomotor. Numeric codes on the following slides are reproduced from Table 8.3.

Source: Criterion 8 SAR, Section 8.2.1 and Table 8.3, pp. 127–130.

HUM 4441: Engineering Ethics

Delivery: Lecture, group discussion, and problem solving.
Assessment: Mid-term and final examinations.

CO	Expected student work (summarized)	Mapped POs	Domains
CO1	Understand moral and ethical principles governing engineering practice and decisions.	PO6	C2
CO2	Examine social and cultural implications, professional responsibility, and accountability.	PO8	C4
CO3	Develop a framework for ethical analysis and engineering decision-making.	PO8	C3
CO4	Analyze and apply ethical implications of data technologies, privacy, security, and ownership.	PO6	C4
CO5	Understand how data-driven technologies create or perpetuate bias and discrimination.	PO6	C2

Source: Criterion 8 SAR, Table 8.3, p. 128; CO descriptions summarized.

CSE 4513: Software Engineering and OOD

Delivery: Lecture and problem solving.
Assessment: Mid-term and final examinations.

CO	Expected student work (summarized)	Mapped POs	Domains
CO1	Explain software development fundamentals and requirement engineering.	PO1, PO2	C2
CO2	Apply design knowledge to model large software systems.	PO2	C3
CO3	Use techniques and tools to release and maintain quality software.	PO1, PO2	C3

Work to explain

Students move from understanding the development process to modeling systems and applying software release and maintenance practices.

Source: Criterion 8 SAR, Table 8.3, p. 128; CO descriptions summarized.

Delivery: Lecture and demonstration.
Assessment: Mid-term and final examinations.

CO	Expected student work (summarized)	Mapped POs	Domains
CO1	Apply general AI techniques to common problem types.	PO1	C3
CO2	Formulate problem scenarios as environments and agents, with justification.	PO2	C5
CO3	Recommend solutions to real-life AI problems using existing techniques.	PO3	C6

Work to explain

Apply AI knowledge $\longrightarrow$ formulate the problem $\longrightarrow$ propose a suitable solution.

Source: Criterion 8 SAR, Table 8.3, p. 128; CO descriptions summarized.

CSE 4712: Artificial Intelligence Lab

Delivery: Group discussion, demonstration, and problem solving.
Assessment: Lab viva, lab tasks, and lab reports.

CO	Expected student work (summarized)	Mapped POs	Domains
CO1	Implement and analyze algorithms for complex navigation and strategy problems.	PO5	C3, C4
CO2	Formulate complex real-world scenarios as specific AI paradigms.	PO2	C5
CO3	Design, test, and evaluate intelligent agents under uncertainty using modern tools and frameworks.	PO4	C6, P5
CO4	Communicate analysis, design choices, and performance through structured technical reports.	PO10	P7, C5

The PO mappings here follow Table 8.3; Table 8.4 differs for CO1 and CO3 (see the review slide).

Source: Criterion 8 SAR, Table 8.3, pp. 128–129; CO descriptions summarized.

CSE 4610: Design Project

Delivery: Group discussion, demonstration, and project work.
Assessment: Presentation and project.

CO	Expected student work (summarized)	Mapped POs	Domains
CO1	Identify research problems and implement innovative solutions with current technologies, tools, and algorithms.	PO1–PO5	C4
CO2	Demonstrate teamwork, ethics, responsibility, and project management to deliver a software system within constraints.	PO8, PO9, PO11	C3, A4
CO3	Design and evaluate computing solutions considering ethical, societal, and environmental implications.	PO6, PO7, PO8	C5, A3
CO4	Communicate outcomes through a structured research report and formal presentation.	PO10	C3, A3
CO5	Build a foundation for future research in the same or related domains.	PO12	C2, A2

Source: Criterion 8 SAR, Table 8.3, p. 129; CO descriptions summarized.

CSE 4700: Project/Thesis

Delivery: Discussion, demonstration, problem solving, and report writing.
Assessment: Presentation and report.

CO	Expected student work (summarized)	Mapped POs	Domains
CO1	Identify and formulate a significant research problem through literature review and analysis of existing solutions.	PO1, PO2, PO4	C4, A4
CO2	Propose methodologies, models, or systems for the identified problem.	PO3	C6, A5, P5
CO3	Conduct experiments, simulations, or data-driven investigations; evaluate solutions and interpret results.	PO4, PO5	C4, A4, P4
CO5	Prepare a technical report and present findings for technical and non-technical audiences.	PO10	C5, P5, A7
CO6	Work independently and collaboratively, apply project management, and pursue continuous learning.	PO9, PO11, PO12	C6, P5, A7

CO numbering follows the source: Table 8.3 does not list a CO4 for this course.

Source: Criterion 8 SAR, Table 8.3, pp. 129–130; CO descriptions summarized.

CSE 4800: Project/Thesis

Delivery: Discussion, demonstration, problem solving, and report writing.
Assessment: Presentation and report.

CO	Expected student work (summarized)	Mapped POs	Domains
CO2	Propose methodologies, models, or systems for the identified problem.	PO3	C6, A5, P5
CO3	Conduct experiments, simulations, or data-driven investigations; evaluate solutions and interpret results.	PO4, PO5	C4, A4, P4
CO4	Demonstrate ethical responsibility and professionalism in research, considering society and the environment.	PO6, PO7, PO8	C6, P6
CO5	Prepare a technical report and present findings for technical and non-technical audiences.	PO10	C5, P5, A7
CO6	Work independently and collaboratively, apply project management, and pursue continuous learning.	PO9, PO11, PO12	C6, P5, A7

CO numbering follows the source: Table 8.3 does not list a CO1 for this course.

Source: Criterion 8 SAR, Table 8.3, p. 130; CO descriptions summarized.

8.2.2: How the courses collectively contribute to POs

Course	PO coverage derived from the CO statements in Table 8.3
HUM 4441: Engineering Ethics	PO6, PO8
CSE 4513: Software Engineering and OOD	PO1, PO2
CSE 4711: Artificial Intelligence	PO1, PO2, PO3
CSE 4712: Artificial Intelligence Lab	PO2, PO4, PO5, PO10
CSE 4610: Design Project	PO1–PO12 (all twelve)
CSE 4700: Project/Thesis	PO1, PO2, PO3, PO4, PO5, PO9, PO10, PO11, PO12
CSE 4800: Project/Thesis	PO3, PO4, PO5, PO6, PO7, PO8, PO9, PO10, PO11, PO12

Why the distribution matters

Theory develops knowledge and analysis; laboratory work adds implementation and investigation; projects integrate technical and professional capabilities.

Source: Criterion 8 SAR, Section 8.2.2 and Tables 8.3–8.4, pp. 128–131; coverage uses Table 8.3 where the tables differ.

Source review: reconcile the CO–PO tables

Table 8.3 and Table 8.4 contain differences that should be checked before finalizing the SAR or presentation.

Course / CO	Table 8.3: statements	Table 8.4: mapping matrix
CSE 4712 / CO1	PO5	PO4
CSE 4712 / CO3	PO4	PO3
CSE 4700 / CO5	PO10	PO6
CSE 4800 / CO5	PO10	No PO marked

Treatment in this draft

Course slides and the coverage summary consistently use Table 8.3. Retain the original CO numbering and learning-domain codes pending confirmation against the approved course outlines.

Source: Criterion 8 SAR, Table 8.3, pp. 128–130; Table 8.4, p. 131.

What Sections 8.1 and 8.2 document

1. **Graduate expectations:** twelve defined POs, with an approval reference.
2. **Program alignment:** PO–PEO mappings and a rationale for every relationship.
3. **Course design:** CO statements, learning domains, delivery methods, and assessment tools for seven selected core courses.
4. **Review responsibilities:** instructor preparation, coordinator harmonization, departmental review, and formal approval of significant changes.
5. **Traceability:** each course outcome connects student work to one or more graduate outcomes.

Presentation takeaway

These sections explain how the program defines its outcomes and organizes course work to support them.

Source: Criterion 8 SAR, Sections 8.1–8.2, pp. 125–131.